

Project Chimera — Architecture Strategy

Document: `architecture_strategy.md`

1. Executive Summary

Project Chimera is a platform that manufactures and operates thousands of Autonomous AI Influencers simultaneously. Each influencer researches trends, generates content, manages social media engagement and participates in commerce.

This document defines:

- The Agent Pattern chosen and why
 - Where humans intervene (Human-in-the-Loop)
 - The database strategy for high-velocity video metadata
 - The full system architecture with diagrams
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2. Agent Pattern Decision

2.1 Patterns Considered

Hierarchical Swarm

2.2 Why Hierarchical Swarm

Project Chimera requires:

- Thousands of influencers running simultaneously
- Multiple content tasks executing in parallel per influencer
- Clear authority chains for quality control
- Independent engagement handling alongside content creation

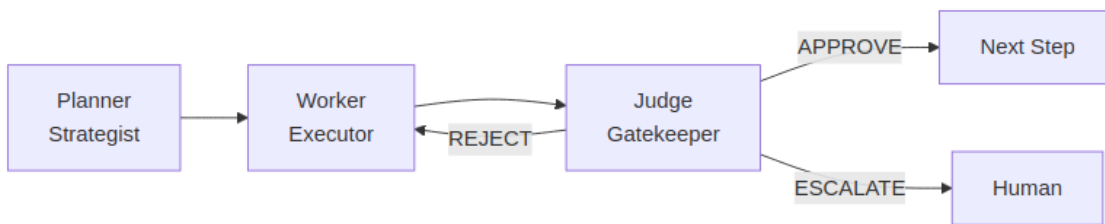
A **Hierarchical Swarm** satisfies all of these because:

1. A top-level Orchestrator commands Campaign Managers
2. Campaign Managers command Planners
3. Planners command Workers
4. All parallel threads operate independently
5. Judges at every layer enforce quality without blocking other threads

A Sequential Chain would mean one task finishes before the next starts - unacceptable at the scale of thousands of influencers and millions of daily operations.

2.3 The Three-Role Core (FastRender Swarm Pattern)

Every campaign thread runs on three specialised roles:

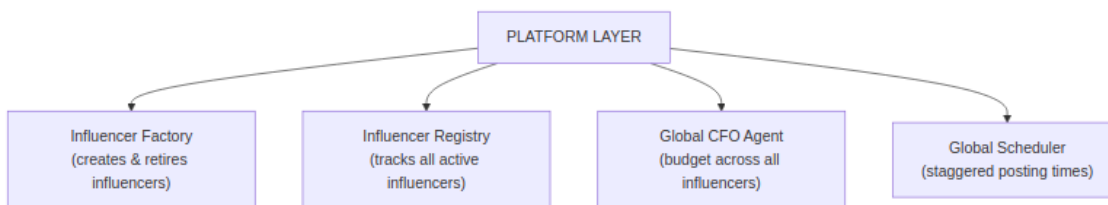


- **Planner:** Thinks and strategises. Breaks goals into tasks.
- **Worker:** Executes one atomic task with maximum speed.
- **Judge:** Reviews output. Has absolute authority to approve, reject, or escalate.

3. Full System Architecture

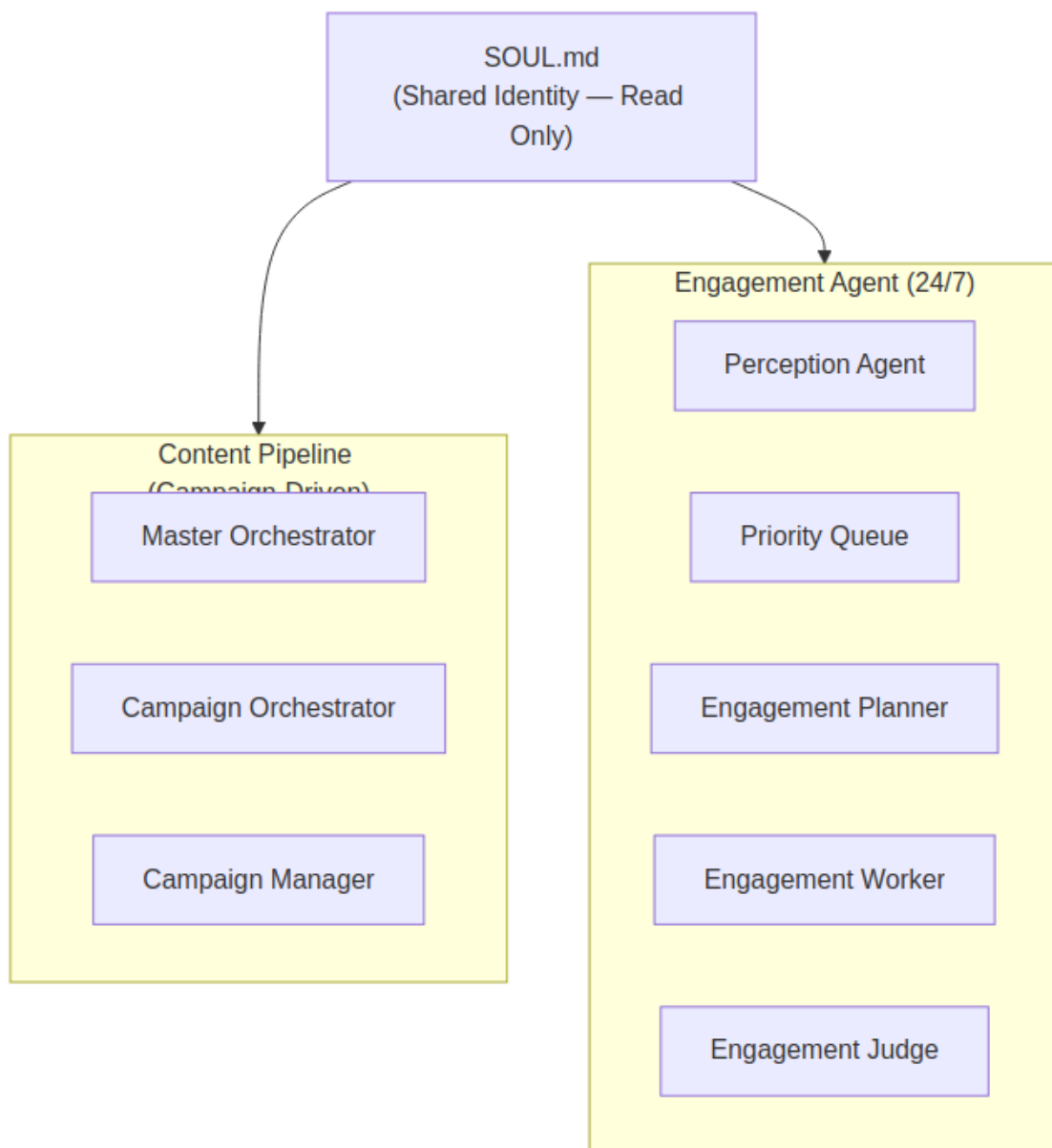
3.1 Platform Layer (Top Level)

The Platform Layer sits above all individual influencers. It manages the fleet.

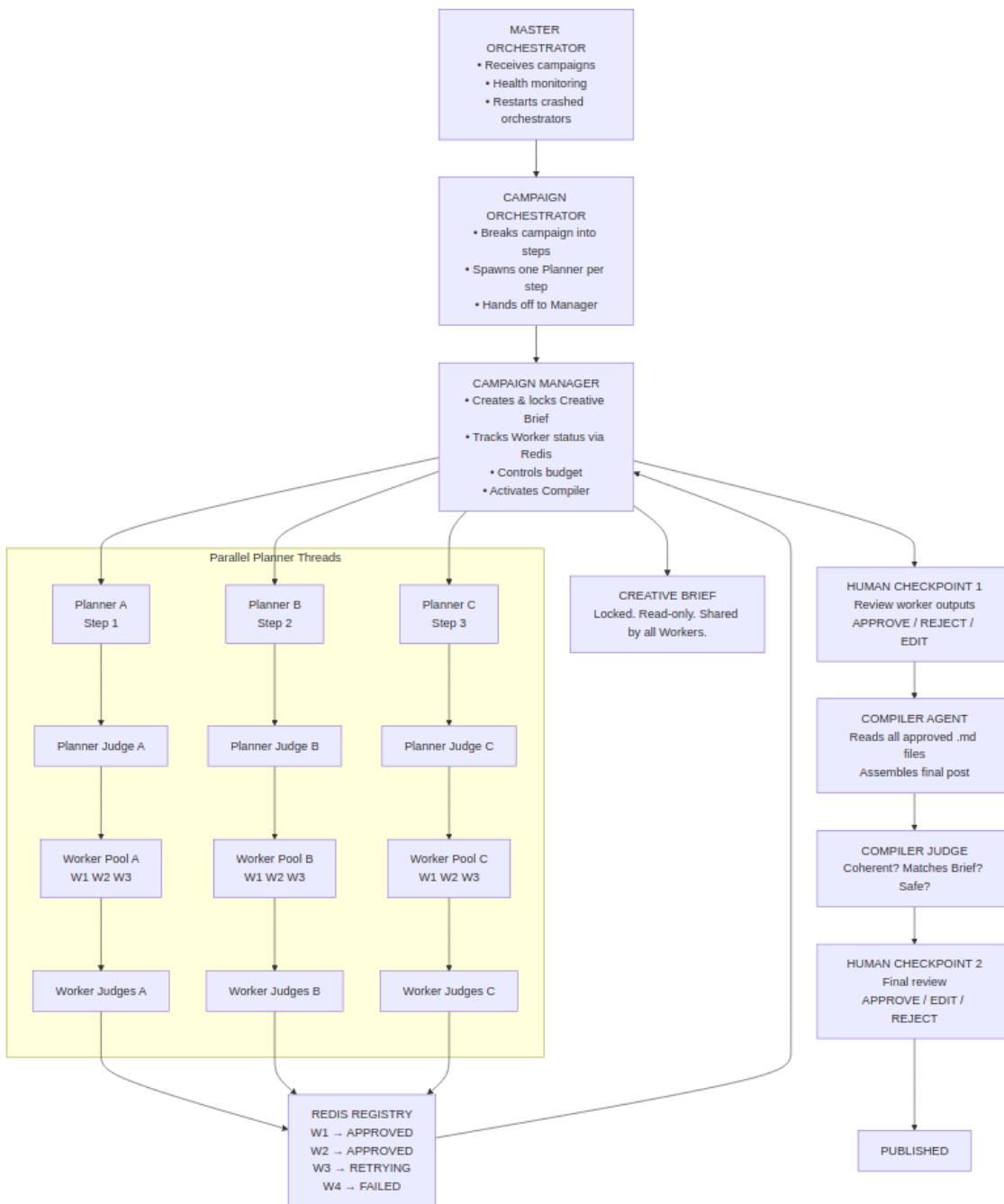


3.2 Per-Influencer Architecture

Each influencer runs two completely independent systems that share one identity file: `SOUL.md`.

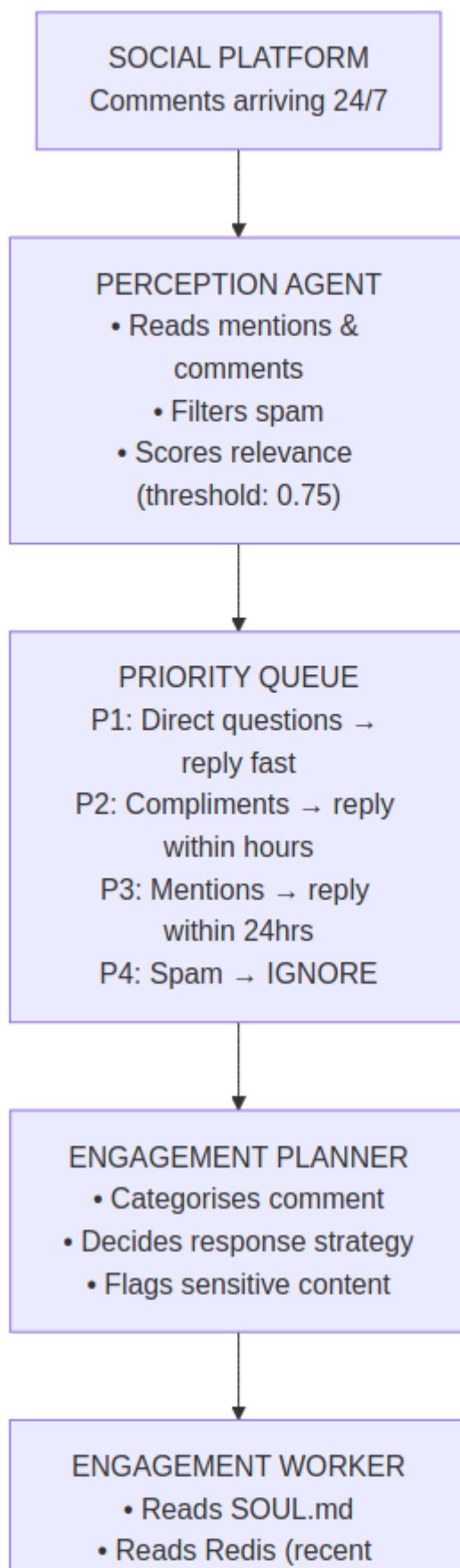


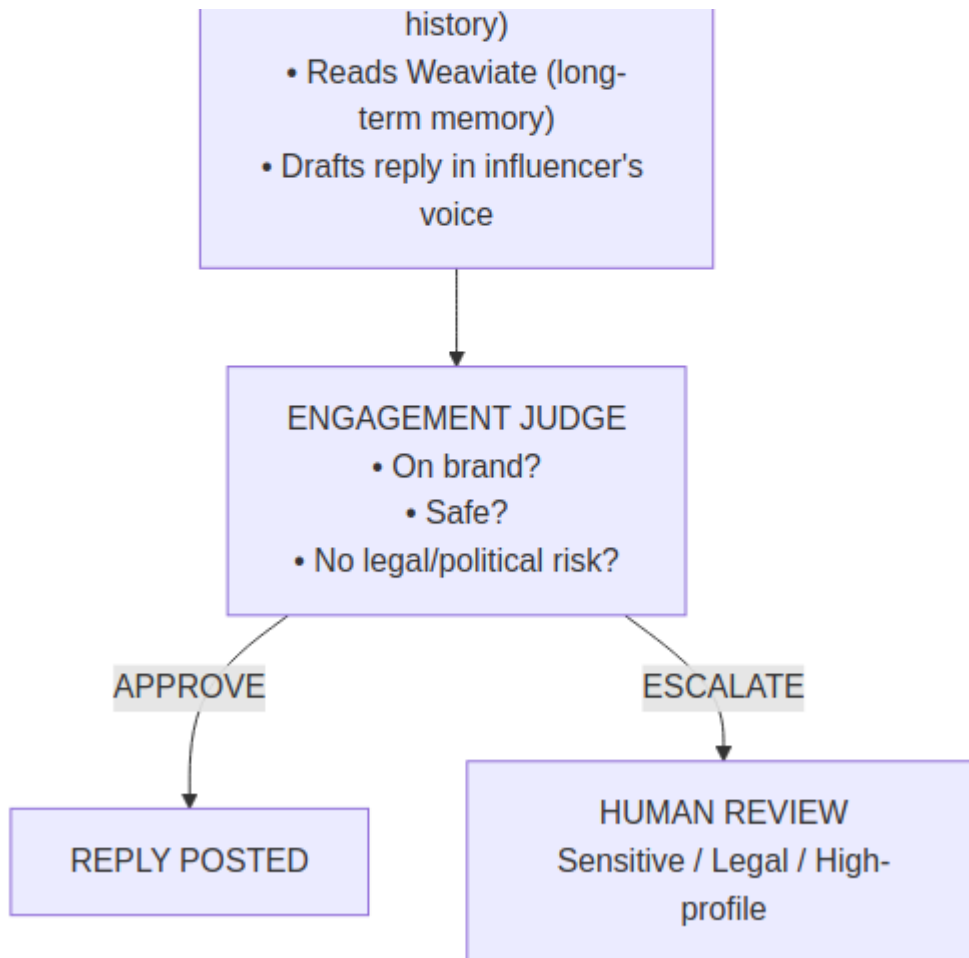
3.3 Content Pipeline — Full Flow



3.4 Engagement Agent — Full Flow

Runs independently from the Content Pipeline. Never blocked by campaigns.



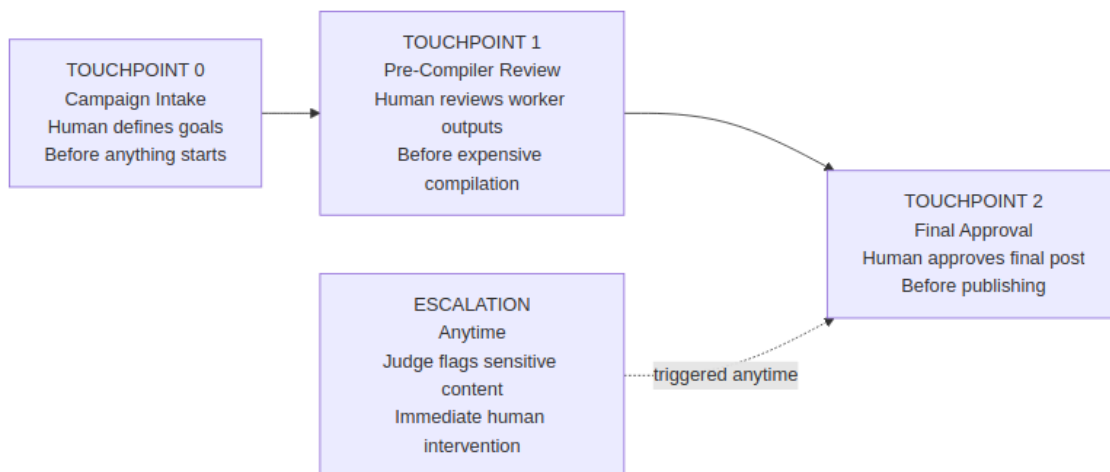


4. Human-in-the-Loop (HITL) Strategy

4.1 Philosophy

Humans do not run the system. They **govern** it. The system operates autonomously unless it reaches a checkpoint or escalation trigger.

4.2 The Three Human Touchpoints



Touchpoint	When	Human Action	Cost of Getting it Wrong
0 - Campaign Intake	Before pipeline starts	Define goals, tone, budget	None - nothing has run yet
1 - Pre-Compiler	After all Workers approved	Approve / Reject / Edit outputs	Low - just redo specific Workers
2 - Final Approval	After Compiler Judge approves	Approve / Edit / Reject final post	Medium - redo compilation
Escalation	Anytime, triggered by Judge	Handle sensitive content	Prevents brand/legal damage

4.3 Escalation Triggers — What Always Goes to a Human

Trigger	Reason
Legal threats in comments	Liability
Harassment	Safety
Political topics	Brand risk
Verified / high-profile account interaction	High visibility mistake
Budget limit exceeded	Financial control

Worker fails 3 retries	Quality breakdown
Low confidence score from Judge	Uncertainty
Impersonation claims	Legal risk

5. Database Strategy

5.1 The Core Question

High-velocity video metadata means:

- Hundreds of videos processed simultaneously
- Metadata structure varies per video (some have transcripts, some don't)
- Constant updates (views, likes, engagement scores)
- Need to query by semantic similarity ("find videos like this one")

This rules out a single database solution. **Chimera uses a polyglot persistence strategy** — the right database for each type of data.

5.2 Database Decision Matrix

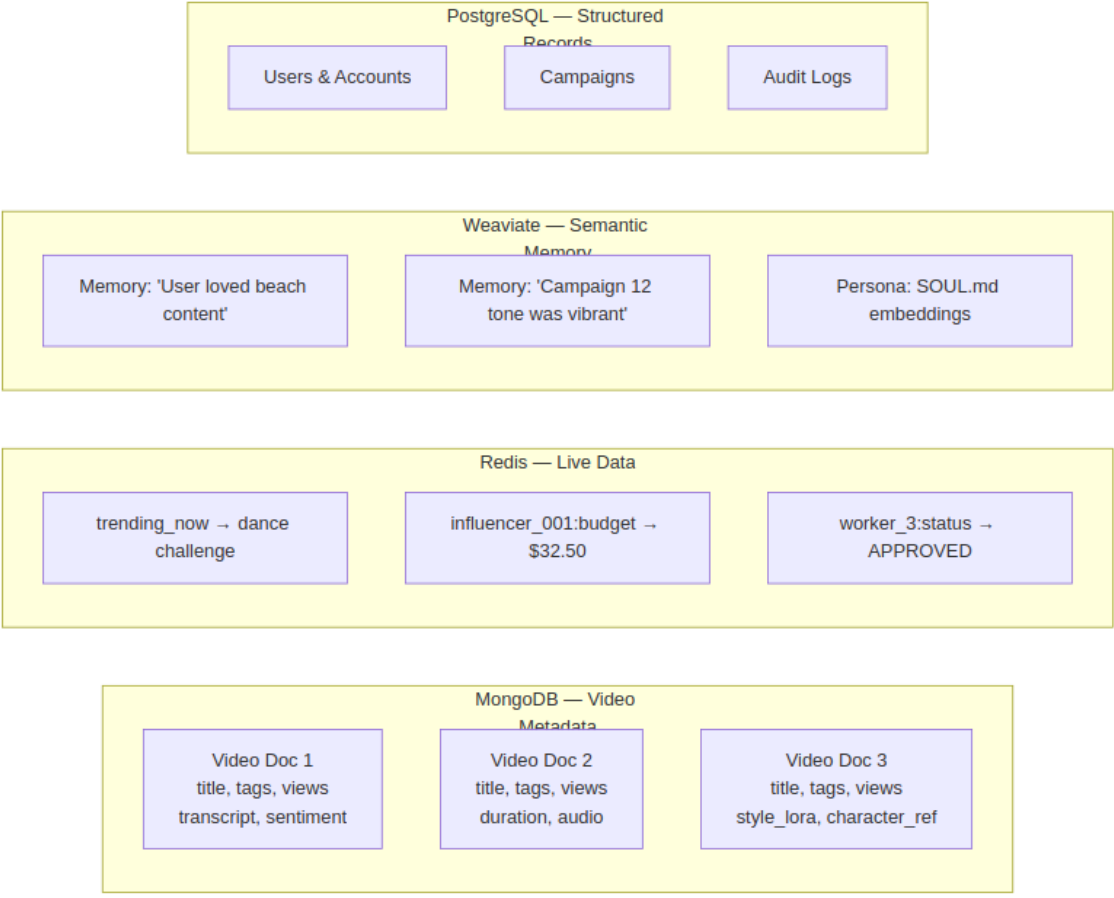
Data Type	Volume	Structure	Best Fit	Choice
Video metadata	Very High	Flexible, varies	Document Store	MongoDB
Trending topics, live counters	Extreme	Key-Value	In-memory cache	Redis
Agent memories, semantic search	High	Vector embeddings	Vector DB	Weaviate
Users, campaigns, financial logs	Medium	Fixed, relational	SQL	PostgreSQL
Financial transactions	Low	Immutable ledger	Blockchain	Base / Ethereum

5.3 Why MongoDB for Video Metadata

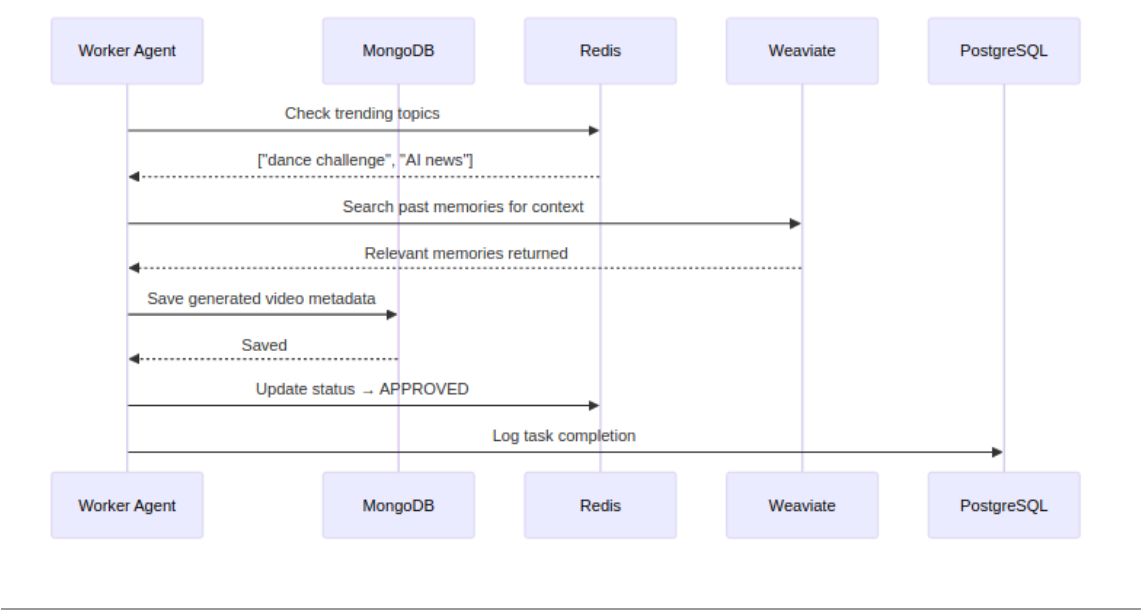
SQL forces every row to have the same columns. Video metadata does not cooperate:

```
Video A: { title, tags, views, transcript, sentiment_score }
Video B: { title, tags, views, duration, audio_fingerprint, location }
Video C: { title, tags, views, captions, character_reference_id, style_lora }
```

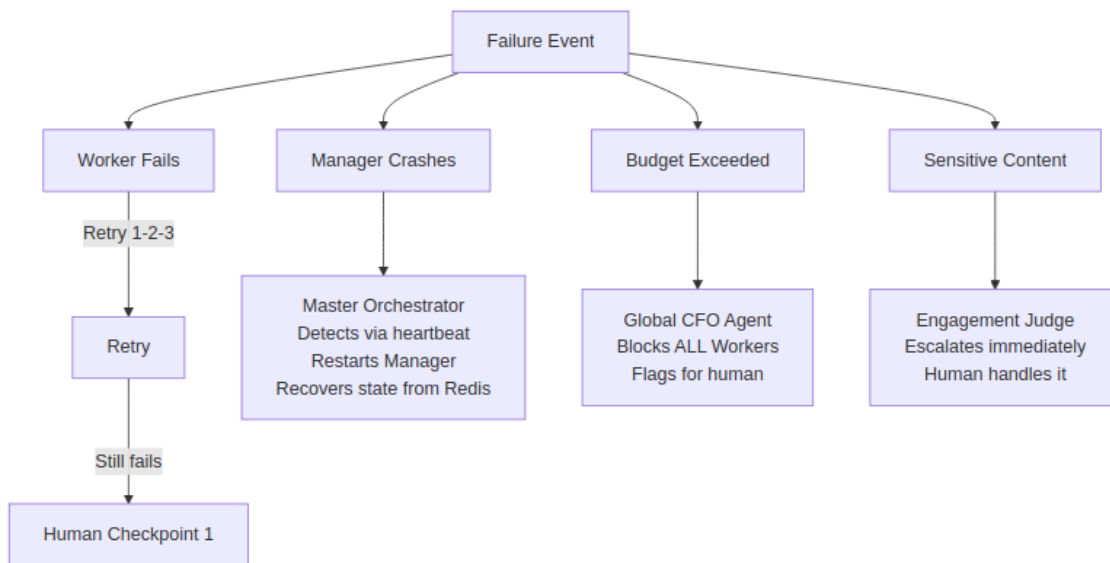
MongoDB stores each as a flexible document. No empty columns. No schema migrations every time a new field is needed.



5.4 Data Flow Through the Stack



6. Failure Handling Summary



7. Technology Stack Summary

Layer	Technology	Purpose
Language	Java 21+	Virtual Threads for massive concurrency
Framework	Spring Boot	Enterprise agent services
Agent Protocol	A2A (Agent-to-Agent)	Inter-agent communication
External Tools	MCP (Model Context Protocol)	Agent-to-tool communication
Content Pipeline State	Redis	Task queue, status registry
Video Metadata	MongoDB	Flexible high-velocity document storage
Long-term Memory	Weaviate	Semantic vector search
Structured Data	PostgreSQL	Users, campaigns, logs
Financial Ledger	Base / Ethereum	Immutable transaction records
AI Reasoning	Claude Opus / Gemini Pro	Complex planning and judging
AI Routine Tasks	Claude Haiku / Gemini Flash	High-volume low-cost tasks
Containers	Docker + Kubernetes	Auto-scaling agent workloads
Cloud	AWS / GCP Hybrid	High availability infrastructure

8. Architecture in One Paragraph

Project Chimera uses a **Hierarchical Swarm pattern** where a Platform Layer manages thousands of independent Influencer Instances. Each instance runs two parallel systems: a Campaign-driven Content Pipeline and a 24/7 Engagement Agent, both sharing a single identity file (`SOUL.md`). The Content Pipeline flows from Master Orchestrator → Campaign Orchestrator → Campaign Manager → parallel Planners → parallel Workers → three-tier Judges (Planner, Worker, Compiler) → two Human Checkpoints → Published. The Engagement Agent runs independently, filtering comments through a Priority Queue, drafting replies using the influencer's memory and voice, and escalating sensitive content to humans immediately. The database strategy uses MongoDB for flexible high-velocity video metadata, Redis for live state and caching, Weaviate for semantic agent memory, and PostgreSQL for structured relational data. Humans intervene at three defined points and via escalation — governing the system without running it.